

TMFC043 Fault Management v1.0.0

Team Approved

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Notice

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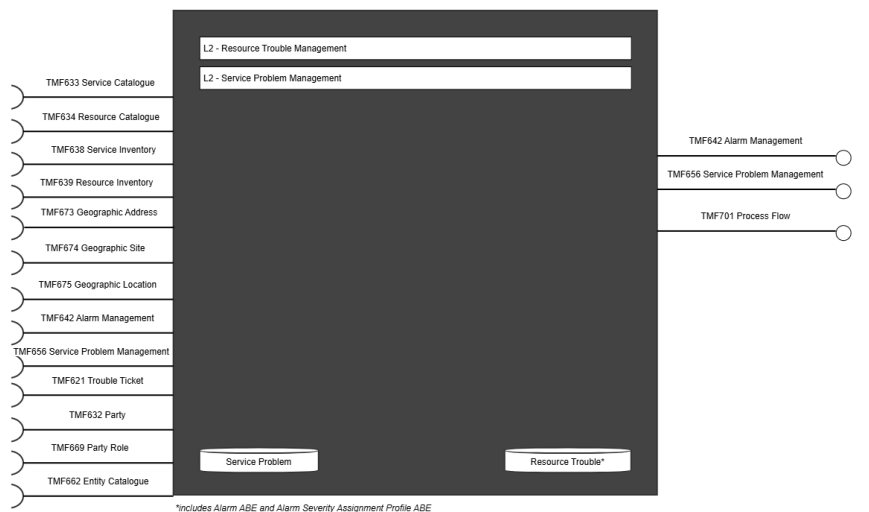
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1. Overview

Component Name	ID	Description	ODA Function Block
Fault Management	TMF C043	Fault management involves identifying, diagnosing, and resolving issues that prevent the Production resources and their overlaying services from functioning correctly. It s scope covers this functionality at the ODA Production layer. The Fault Management component manages the active alarms or problem candidates. Potentially modifying or enhancing their data to make the persistent alarms more actionable. It may apply logic to preserve alarms that reflect actual problems, eliminating those that are merely symptoms. Various actions can be taken using the alarm/problem information as part of the resolution process, such as Production applying automated corrective actions or opening trouble tickets.	Production

TMF043: Fault Management



2. eTOM Processes, SID Data Entities and Functional Framework Functions

2.1. eTOM business activities

eTOM business activities this ODA Component is responsible for.

Identifier	Level	Business Activity Name	Description
1.5.8	L2	Resource Trouble Management	Resource Trouble Management processes are responsible for the management of troubles, including security events, associated with specific resources. The objectives of these processes are to efficiently and effectively manage reported resource trouble, isolate the root cause and act to resolve the resource trouble.
1.4.6	L2	Service Problem management	Service Problem Management processes are responsible for the management of problems associated with specific services. The objective of these processes is to respond immediately to reported service problems or failures in order to minimize their effects on customers, and to invoke the restoration of the service, or provide an alternate service as soon as possible.

2.2. SID ABEs

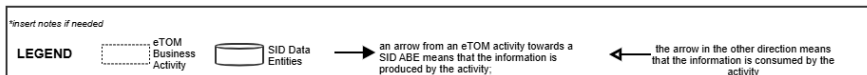
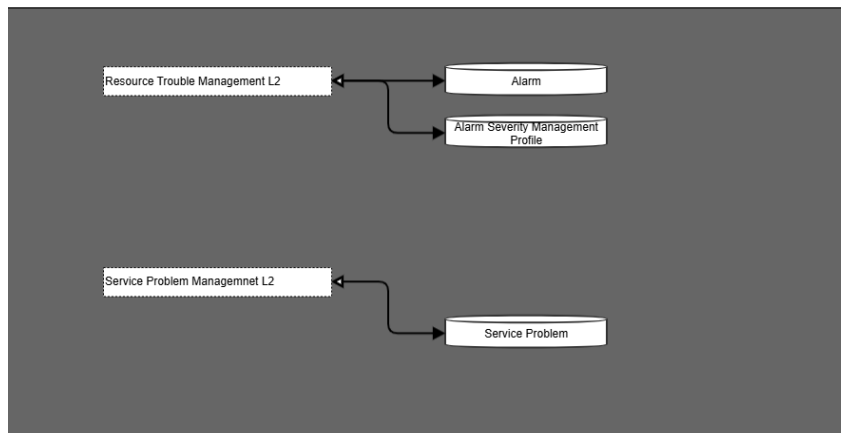
SID ABEs this ODA Component is responsible for:

SID ABE Level 1	SID ABE Level 2 (or set of BEs)*
Resource Trouble ABE	Resource Alarm ABE
Resource Trouble ABE	Alarm Severity Assignment Profile ABE
Service Problem ABE	

*: if SID ABE Level 2 is not specified this means that all the L2 business entities must be implemented, else the L2 SID ABE Level is specified.

2.3. eTOM L2 - SID ABEs links

eTOM L2 vS SID ABEs links for this ODA Component.



2.4. Functional Framework Functions

Function ID	Function Name	Function Description	Aggregate Function Level 1	Aggregate Function Level 2
5.6.2.4	Resource Trouble Analytics	Resource Trouble Cause Identification Function supports the trouble causes identification, that is to say the reasons why the installed resource is in trouble and among these reasons the root cause of the resource trouble.	Resource Trouble Management	Resource Trouble Analytics
5.6.2.2	Resource Trouble Resolution	Resource Trouble Resolution Function identifies the necessary resolution actions to solve the Resource Trouble according to Resource Problem Policy. It is based on diagnostic results.	Resource Trouble Management	Resource Trouble Resolution
4.6.1	Service Problem Management	Service Problem Management groups functions in charge of managing Service Trouble.	Service Problem Management	

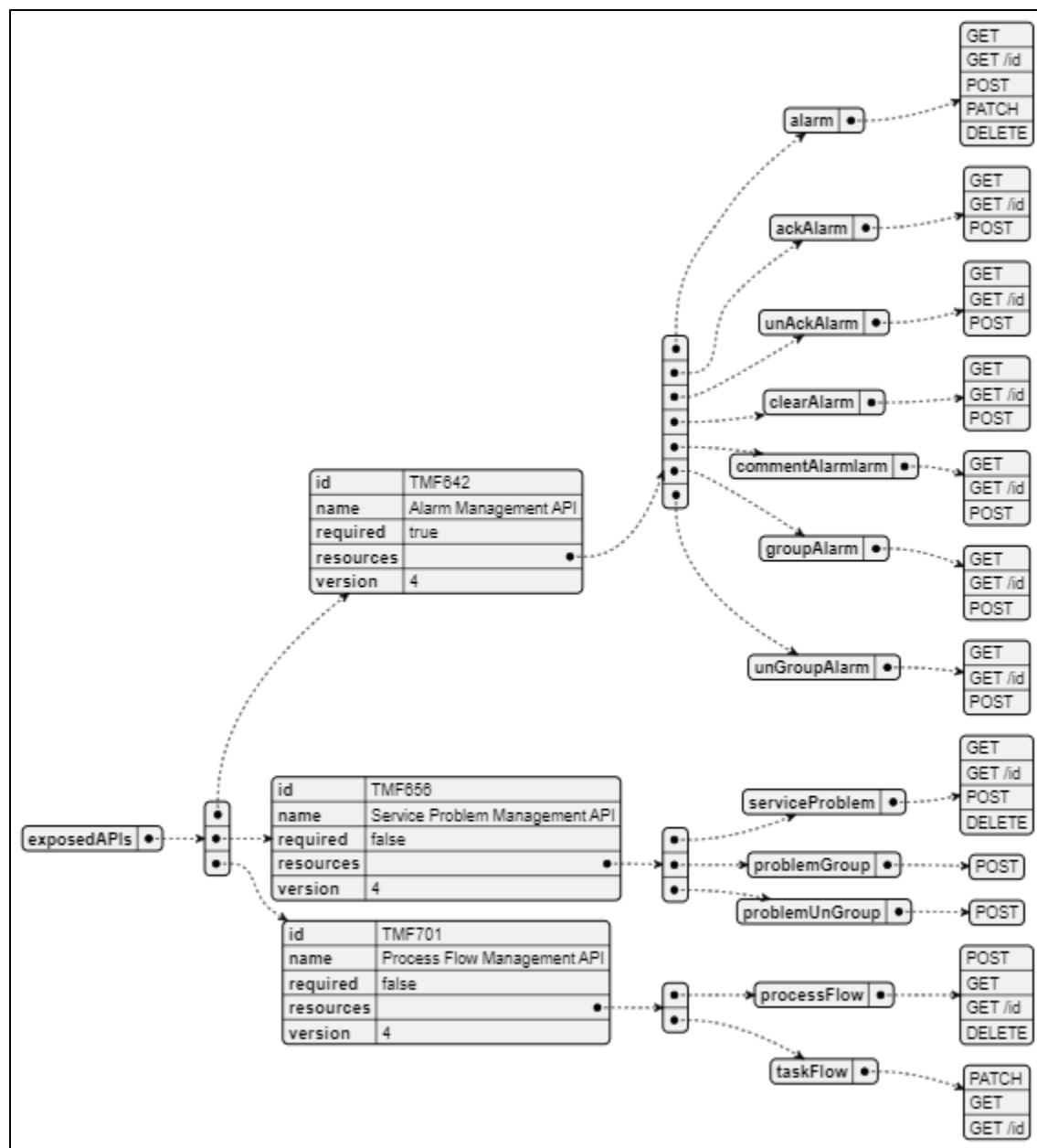
3. TM Forum Open APIs & Events

The following part covers the APIs and Events; This part is split in 3:

- List of **Exposed APIs** - This is the list of APIs available from this component.
- List of **Dependent APIs** - In order to satisfy the provided API, the component could require the usage of this set of required APIs.
- List of **Events (generated & consumed)** - The events which the component may generate is listed in this section along with a list of the events which it may consume. Since there is a possibility of multiple sources and receivers for each defined event.

3.1. Exposed APIs

Following diagram illustrates API/Resource/Operation:

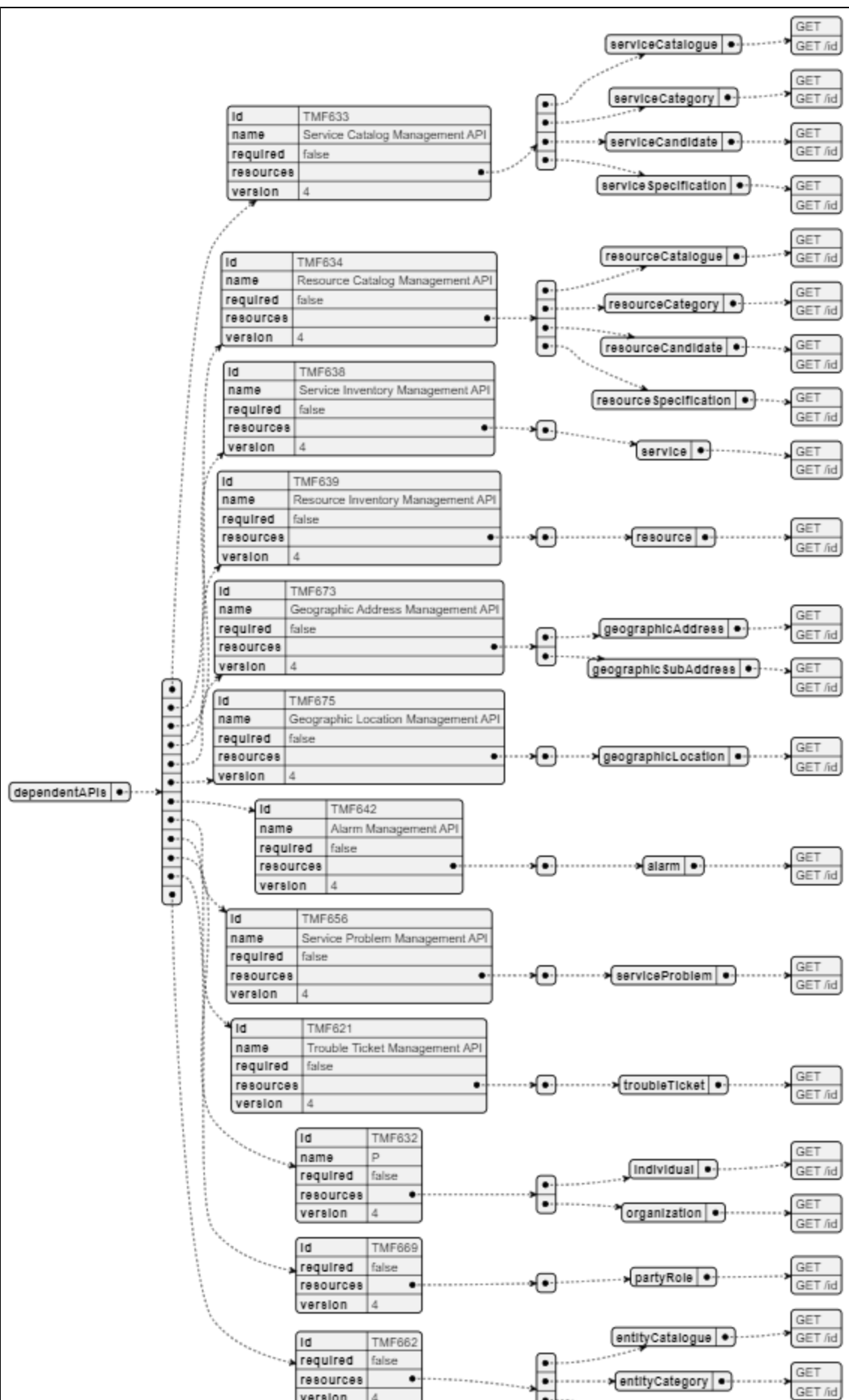


API ID	API Name	API Version	Mandatory / Optional	Operations	Rationale
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TMF642	Alarm Management	4	mandatory	alarm: - POST - GET - GET /id - PATCH /id - DELETE /id AckAlarm: (optional) ◦ POST ◦ GET ◦ GET /id UnAckAlarm: (optional) ◦ POST ◦ GET ◦ GET /id ClearAlarm: ◦ POST ◦ GET ◦ GET /id CommentAlarm: (optional) ◦ POST ◦ GET ◦ GET /id GroupAlarm: (optional) ◦ POST ◦ GET ◦ GET /id UnGroupAlarm: (optional) ◦ POST ◦ GET ◦ GET /id	
TMF656	Service Problem Management	4	optional	serviceProblem: - POST - GET - GET /id - PATCH /id - DELETE /id serviceProblemEventRecord:(optional) ◦ GET ◦ GET /id problemGroup:(optional) ◦ Post problemUnGroup:(optional) ◦ Post	
TMF701	Process Flow	4	optional	processFlow: • GET • GET /id • POST • DELETE /id taskFlow: • GET • GET /id • PATCH /id	

3.2. Dependent APIs

Following diagram illustrates API/Resource/Operation potentially used by the Fault Management component:



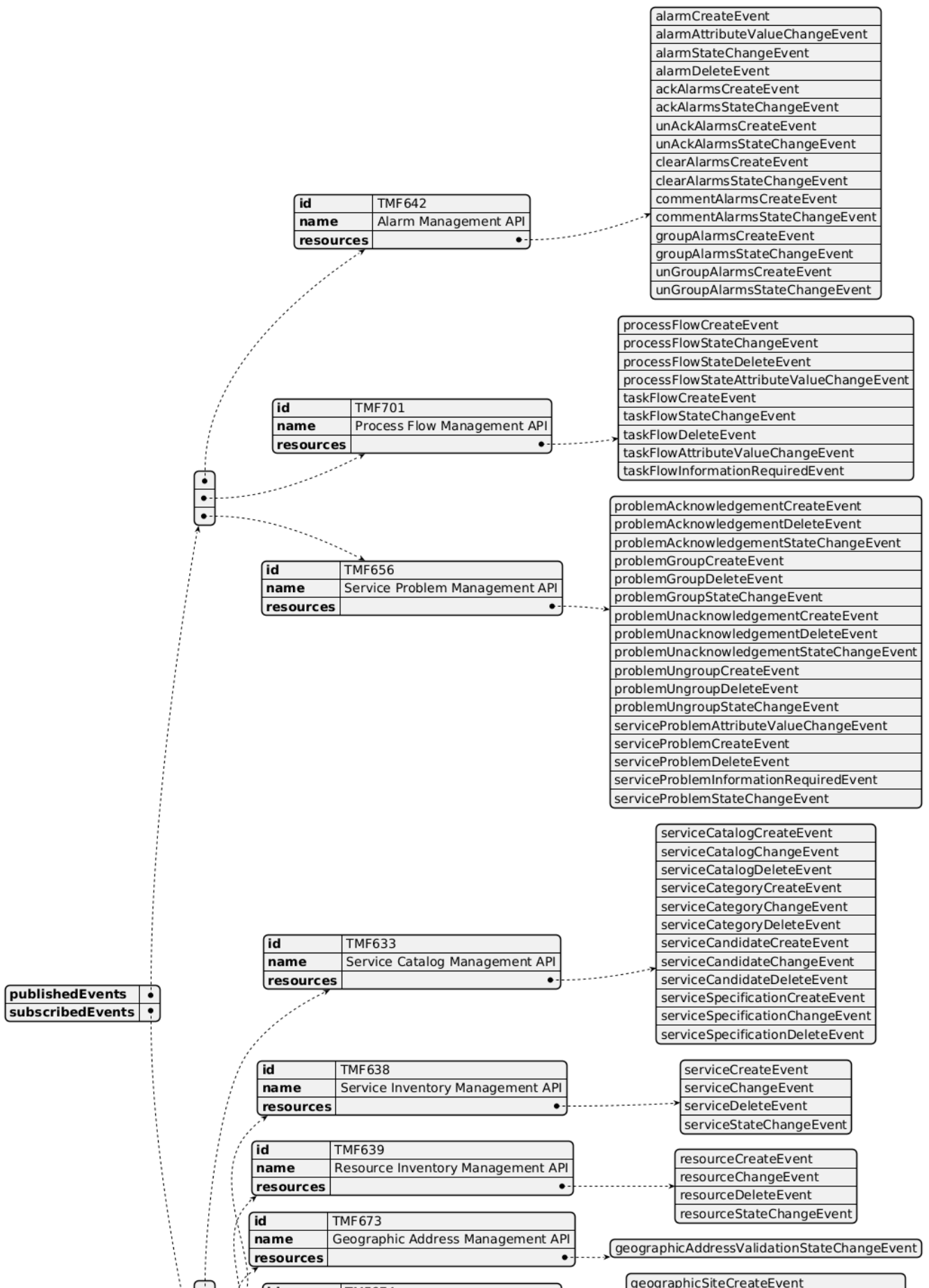


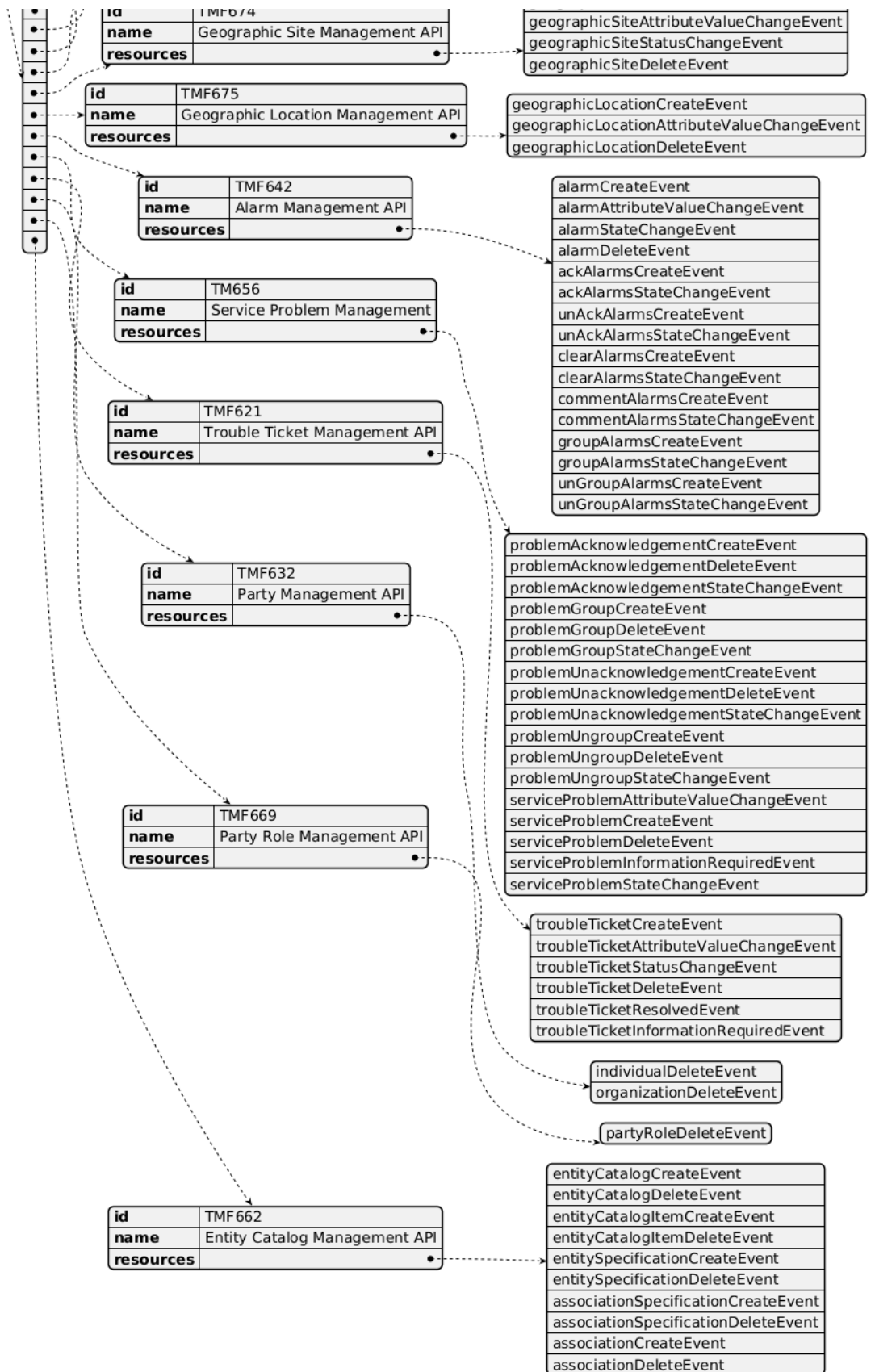
API ID	API Name	API Version	Mandatory / Optional	Operations	Rationale
TMF633	Service Catalogue	4	optional	serviceCatalogue: - GET /id - GET serviceCategory: - GET /id - GET serviceCandidate: - GET /id - GET serviceSpecification: - GET /id - GET	
TMF634	Resource Catalogue	4.1	optional	resourceCatalogue: - GET /id - GET resourceCategory: - GET /id - GET resourceCandidate: - GET /id - GET resourceSpecification: - GET /id - GET	
TMF638	Service Inventory	4	optional	service: - GET /id - GET	
TMF639	Resource Inventory	4	optional	resource: - GET /id - GET	
TMF673	Geographic Address Management	4	optional	geographicAddress: - GET - GET /id geographicSubAddress: - GET - GET /id	
TMF674	Geographic Site	4	optional	geographicSite: - GET - GET /id	
TMF675	Geographic Location	4	optional	geographicLocation: - GET - GET /id retrieveGeographicLocation: - GET - GET /id	
TMF642	Alarm Management	4	optional	alarm: - GET - GET /id	Included to support alarm federation architectures.
TMF656	Service Problem Management	4	optional	serviceProblem: - GET - GET /id	Included to support alarm federation architectures.
TMF621	Trouble Ticket	4	optional	troubleTicket: - GET - GET /id	
TMF632	Party	4	optional	individual: - GET - GET /id organization: - GET - GET /id	The ownership can be achieved by using Party or by using alarm attributes in the alarm ABEmodel
TMF669	Party Role	4	optional	partyRole: - GET - GET /id	

TMF662	Entity Catalogue	4	optional	entityCatalogue: - GET /id - GET entityCategory: - GET /id - GET entityCatalogItem - GET /id - GET entitySpecification: - GET /id - GET	
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3.3. Events

The diagram illustrates the Events which the component may publish and the Events that the component may subscribe to and then may receive. Both lists are derived from the APIs listed in the preceding sections.





4. Machine Readable Component Specification

Refer to the ODA Component table for the machine-readable component specification file for this component.

5. References

5.1. TMF Standards related versions

Standard	Version(s)
SID	24.5
eTOM	24.5
Functional Framework	24.5

5.2. Jira References

5.2.1. SID

[\[ISA-1288\] Missing Network Trouble Ticket Entity - TM Forum JIRA](#)

5.3. Further resources

1. IG1228: please refer to IG1228 for defined use cases with ODA components interactions.

6. Administrative Appendix

6.1. Document History

6.1.1. Version History

Version Number	Date Modified	Modified by:	Description of changes
1.0.0	13-May-2025	Yuval Stein	First Version

6.1.2. Release History

Release Status	Date Modified	Modified by:	Description of changes
Pre-production	13-May-2025	Rosie Wilson	First Version

6.2. Acknowledgements

This document was prepared by the members of the TM Forum Component and Canvas project team:

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